

AB POWER ENGINEERING

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FUEL AND LUBRICANTS

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1. Properties of fuel that affects how easily the fuel flows, particularly at low temperatures, and impacts fuel pump and injector performance.

- a. Specific Gravity
- b. Viscosity
- c. Calorific Value
- d. None of the above

2. The lowest temperature at which a fuel can ignite, indicating its flammability and storage safety.

- a. Flash Point
- b. Calorific Value
- c. Boiling Point
- d. None of the above

3. A measure of a diesel fuel's ignition quality, influencing how easily it burns.

- a. Cetane Number
- b. Octane Number
- c. Calorific Value
- d. None of the above

4. The amount of heat released when a fuel is completely burned, indicating its energy content.

- a. Calorific Value
- b. Flash Point
- c. Pour Point
- d. None of the above

5. This relates fuel volume to weight and indirectly influences energy content.

- a. Specific Gravity
- b. Density
- c. Specific weight
- d. All of the above

6. A measure of a fuel's density relative to water, affecting engine performance and emissions.

- a. Specific Weight
- b. Specific Gravity
- c. Specific Fuel Consumption
- d. All of the above

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7. Determines the lubricant's ability to form a protective film between moving parts and resist flow at different temperatures.

- a. Viscosity
- b. Viscosity Index
- c. Pour Point
- d. None of the above

8. Indicates the temperature at which a lubricant can form an ignitable vapor, affecting safety.

- a. Flash Point
- b. Boiling Point
- c. Pour Point
- d. All of the above

9. Measures a lubricant's ability to neutralize acidic by-products of combustion, indicating its long-term effectiveness.

- a. Base Number
- b. Thermal Stability
- c. Oxidation Resistance
- d. None of the above

10. Indicates how much a lubricant's viscosity changes with temperature, affecting its performance under varying operating conditions.

- a. Viscosity Index
- b. Viscosity
- c. Pour Point
- d. All of the above

11. Refers to a lubricant's resistance to degradation at high temperatures, ensuring its long-term effectiveness.

- a. Flash Point
- b. Pour Point
- c. Thermal Stability
- d. All of the above

12. Measures a lubricant's ability to resist chemical reactions that degrade its properties over time.

- a. Base Number
- b. Thermal Stability
- c. Oxidation Resistance
- d. All of the above

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13. Properties of fuel and lubricants that directly impact how well the engine operates.

- a. Viscosity of fuel
- b. Cetane or octane number of fuels
- c. Viscosity in lubricants
- d. All of the above
- e. Two of the above

13. For fuel economy, proper fuel _____ and other properties contribute to efficient combustion and minimize energy waste.

- a. Viscosity
- b. Specific gravity
- c. Thermal stability
- d. None of the above

14. _____ and other properties like specific gravity help ensure safe handling, storage, and transportation of fuels and lubricants.

- a. Flash point
- b. Pour point
- c. Cloud point
- d. All of the above

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15. Properties that relates to the lubricant effectiveness on how well a lubricant reduces friction, wear, and heat.

- a. Viscosity
- b. Viscosity index
- c. Thermal stability
- d. All of the above

16. Properties that can be easily used to detect if fuels have been adulterated with other substances.

- a. Density
- b. Thermal stability
- c. Calorific value
- d. All of the above

17. Characteristic of good lubricants.

- a. High boiling point
- b. Low freezing point
- c. High viscosity index
- d. Thermal stability
- e. Hydraulic stability.
- f. Corrosion prevention.
- d. High resistance to oxidation
- e. All of the above

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18. The total amount of energy released when a specific amount of fuel is completely burned.

- a. Heating value
- b. Higher heating value
- c. Lower Heating value
- d. None of the above

19. Amount of energy which also includes the latent heat of vaporization of water vapor produced.

- a. Heating value
- b. Higher heating value
- c. Lower Heating value
- d. None of the above

20. This excludes the latent heat of vaporization of moisture from the energy that is available in fuel.

- a. Heating value
- b. Higher heating value
- c. Lower Heating value
- d. None of the above

21. The energy potential of a fuel.

- a. Energy content
- b. Heating value
- c. Calorific value
- d. All of the above

22. It indicate the quality fuel for combustion and improved thermal efficiency.

- a. Combustion efficiency
- b. Fuel efficiency
- c. System efficiency
- d. None of the above

23. The lowest temperature of fuel at which it will flow under its own weight.

- a. Pour point
- b. Cloud point
- c. Flash point
- d. None of the above

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24. Effect of fuel with lower pour point (remain fluid at colder temperatures)

- a. Prevents fuel system blockage
- b. Efficient engine operation
- c. Maintain smooth fuel transfer in cold climates
- d. All of the above

25. Effect of high pour points fuel.

- a. Easily makes the fuel thick and solidify
- b. Lead to engine malfunction
- c. Reduced performance of the engine in cold weather
- d. All of the above

26. The lowest temperature of fuel at which wax crystals begin to form, causing the fuel to appear cloudy.

- a. Pour point
- b. Cloud point
- c. Flash point
- d. None of the above

27. An indicator of Fuel with low Cloud point.

- a. Wax Crystal Formation - dissolved paraffin wax begins to crystallize and solidify
- b. Appearance of Cloudiness - growing crystals scatter light, giving the fuel a cloudy or hazy appearance.
- c. Impact on Fuel Systems - formation of crystals indicate that the fuel is approaching a point where it can no longer flow freely and can lead to fuel filter plugging
- d. All of the above

28. Color code for grease typically use in high temperature or marine applications.

- a. Green
- b. Blue
- c. Red
- d. Black/Brown
- e. White
- f. None of the above

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29. Color code for grease the indicate high performance, biodegradable, or extreme pressure additives for heavy-duty use.

- a. Green
- b. Blue
- c. Red
- d. Black/Brown
- e. White
- f. None of the above

30. Color code for grease typically associate with high-temperature applications or heavy-duty use for automotive and industrial machinery.

- a. Green
- b. Blue
- c. Red
- d. Black/Brown
- e. White
- f. None of the above

31. Color code for grease which is frequently contains moly or graphite designed for extreme load condition and heavy duty applications

- a. Green
- b. Blue
- c. Red
- d. Black/Brown
- e. White
- f. None of the above

32. Color code for grease made of inedible hog fat or other natural ingredients and is often used in applications where food-grade grease is required..

- a. Green
- b. Blue
- c. Red
- d. Black/Brown
- e. White
- f. None of the above

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33. Types of lubricants.

- a. Oil
- b. Grease
- c. Penetrating lubricants
- d. Dry lubricants
- ☒ e. All of the above

34. Lubricants suitable for applications involving grease, hinges, bearings, and other moving parts.

- ☒ a. Oil
- b. Grease
- c. Penetrating lubricants
- d. Dry lubricants
- e. All of the above

35. Lubricants used for applications where oil alone is not suitable such as bearings and chassis components

- a. Oil
- ☒ b. Grease
- c. Penetrating lubricants
- d. Dry lubricants
- e. All of the above

36. They are useful for treating stuck bolts, hinges, and other parts example WD-4D.

- a. Oil
- b. Grease
- ☒ c. Penetrating lubricants
- d. Dry lubricants
- e. All of the above

39. Lubricants suitable for applications where oil or grease would attract dirt or debris or where a clean, dry surface is needed.

- a. Oil
- b. Grease
- ☒ c. Penetrating lubricants
- ☒ d. Dry lubricants
- e. All of the above

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40. An arbitrary, unit-less number that indicates how much a fluid's viscosity (resistance to flow) changes with temperature.

- a. Cloud point
- b. Pour point
- ☒ c. Viscosity index
- d. None of the above

41. If viscosity is high, it means that lubricant is _____ over a period over a broad temperature.

- ☒ a. relatively stable
- b. not relatively stable
- c. not advisable to use
- d. None of the above

42. If viscosity index is low, it means that the lubricant _____ with temperature.

- a. will not change significantly
- ☒ b. will change significantly
- c. is not advisable for use
- d. None of the above

43. What are the effects of high viscosity index lubricants?

- a. Provides better lubrication
- b. Improved fuel efficiency
- ☒ c. Better performance in extreme temperature
- ☒ d. All of the above

44. Allowable storage time for gasoline fuel.

- a. 1 to 2 months
- ☒ b. 3 to 6 months
- c. 7 to 9 months
- d. None of the above

45. Allowable storage time for diesel fuel.

- a. 1 to 5 months
- ☒ b. 6 to 12 months
- c. 13 to 18 months
- d. None of the above

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46. To prolong the shelf life of fuel, it should be stored in _____ containers, away from sunlight and heat.

- a. cool
- b. dry
- c. sealed
- d. airtight
- ☒ e. All of the above

47. Ethanol in gasoline basically is _____ to oxidation and evaporation, reducing its combustibility.

- ☒ a. susceptible
- b. not susceptible
- c. not a problem in
- d. None of the above

48. Diesel degradation factor.

- a. Moisture from condensation
- b. Oxygen exposure
- ☒ c. Temperature fluctuation.
- d. All of the above

49. CO2 emission factor for diesel.

- a. 1.1 to 2.4 kg CO2 per liter
- ☒ b. 2.5 to 2.7 kg CO2 per liter
- c. 2.8 to 3.1 kg CO2 per liter
- d. None of the above

50. CO2 emission factor for gasoline.

- ☒ a. 0.63 kg CO2 per gallon
- b. 0.80 kg CO2 per gallon
- c. 1.1 kg CO2 per gallon
- d. None of the above

51. Which of the following are considered fuel emission of gasoline?

- a. Greenhouse gases (CO2, CH4, and N2O)
- b. Climate change and air pollutants (CO, NOx)
- c. Unburned hydrocarbons
- d. Particulate matters
- ☒ e. All of the above

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52. Heating value of gasoline fuel. (1 MJ x 239 kcal)

- ☒ a. 45.6 MJ/kg
- b. 50.1 MJ/kg
- c. 59.5 MJ/kg
- d. None of the above

53. Heating value of diesel fuel. (1 MJ x 239 kcal)

- ☒ a. 43.1 MJ/kg
- b. 46.7 MJ/kg
- c. 50.1 MJ/kg
- d. None of the above

54. Stoichiometric air requirement of gasoline fuel.

- ☒ a. 14.7 kg fuel/kg air
- b. 20.1 kg fuel/kg air
- c. 25.3 kg fuel/kg air
- d. None of the above

55. Stoichiometric air requirement of diesel fuel.

- ☒ a. 14.5 kg fuel/kg air
- b. 19.2 kg fuel/kg air
- c. 21.4 kg fuel/kg air
- d. None of the above

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56. The supply of more air than the theoretical amount needed for complete combustion.

- a. Primary air
- b. Secondary air
- ☒ c. Excess air
- d. None of the above

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GOD BLESS!!!

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